

Lab 06: OpenMP Parallel Programming

Exercise 02 Report: OpenMP Single Directive

1. Exercise Goal

Analyze the behavior of `#pragma omp single` and its implicit barrier.

2. Source Code

```
#include <omp.h>
#include <stdio.h>

int main() {
    #pragma omp parallel
    {
        int th = omp_get_thread_num();
        printf("thread # %d\n", th);

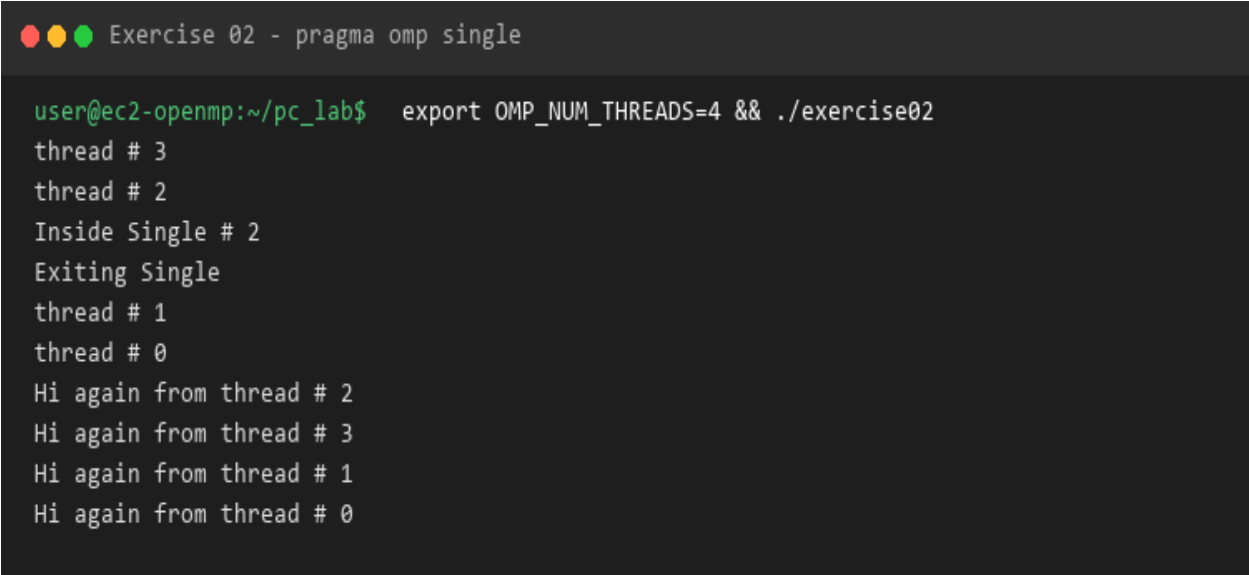
        #pragma omp single
        {
            printf("Inside Single # %d\n", omp_get_thread_num());
            printf("Exiting Single \n");
        }
        printf("Hi again from thread # %d\n", omp_get_thread_num());
    }
    return 0;
}
```

3. Explanation & Analysis

The `#pragma omp single` directive specifies that the enclosed block is executed by only one thread in the team (whichever thread reaches it first). The other threads in the team skip executing the single section.

By default, there is an implicit barrier at the end of the single block, so all threads wait for the thread executing the single block to finish before continuing.

5. Execution Output Screenshot



```
Exercise 02 - pragma omp single

user@ec2-openmp:~/pc_lab$ export OMP_NUM_THREADS=4 && ./exercise02
thread # 3
thread # 2
Inside Single # 2
Exiting Single
thread # 1
thread # 0
Hi again from thread # 2
Hi again from thread # 3
Hi again from thread # 1
Hi again from thread # 0
```